

```
In [1]: import matplotlib.pyplot as plt
```

```
In [2]: import pandas as pd
```

```
In [3]: import numpy as np
```

```
In [5]: x = np.random.randint(100,size=(25))
```

```
In [6]: y = np.random.randint(100,size=(25))
```

```
In [7]: x
```

```
Out[7]: array([16,  1,  5, 79, 86, 43, 25, 16, 17,  9, 38, 81, 20, 13, 27, 55, 14,
              44, 78, 47, 21, 81, 58,  5,  9])
```

```
In [8]: y
```

```
Out[8]: array([14, 81, 21, 74, 27,  1, 27, 56, 75, 44, 78, 56, 45, 43, 51,  2, 22,
              35, 71, 25, 74, 70, 86, 63, 64])
```

```
In [11]: from sklearn.linear_model import LinearRegression
```

```
In [12]: lr1 = LinearRegression()
```

```
In [22]: X_train = x.reshape(-1,1)
```

```
In [23]: Y_train = y.reshape(-1,1)
```

```
In [24]: lr1.coef_
```

```
Out[24]: array([[0.06263421]])
```

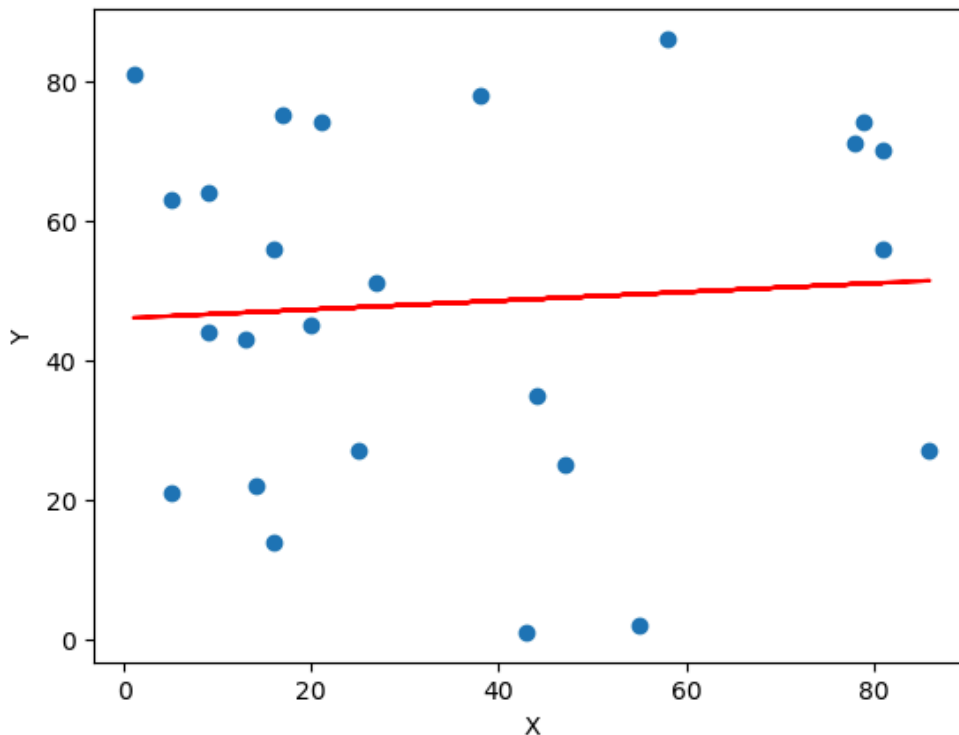
```
In [25]: lr1.intercept_
```

```
Out[25]: array([45.97523291])
```

```
In [26]: y_pred = lr1.predict(X_train)
```

```
In [32]: plt.scatter(x,y)
plt.xlabel("X")
plt.ylabel("Y")
plt.plot(X_train, y_pred, color='red')
```

```
Out[32]: [<matplotlib.lines.Line2D at 0x1c7fe63df10>]
```



```
In [36]: from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
In [37]: mean_absolute_error(Y_train, y_pred)
```

```
Out[37]: 21.545271325787716
```

```
In [38]: mean_squared_error(Y_train, y_pred)
```

```
Out[38]: 623.4652115254643
```

```
In [39]: r2_score(Y_train, y_pred)
```

```
Out[39]: 0.004685166785657335
```

```
In [40]: df = pd.read_csv('placement - placement.csv')
```

```
In [41]: X = df['cgpa']
```

```
In [70]: X = X.values.reshape(-1,1)
```

In [71]:

```
X
[7.05],
[7.62],
[6.66],
[6.13],
[6.33],
[7.76],
[7.77],
[8.18],
[5.42],
[8.58],
[6.94],
[5.84],
[8.35],
[9.04],
[7.12],
[7.4 ],
[7.39],
[5.23],
[6.5 ],
[5.12],
```

In [52]: `y = df['package']`

In [53]:

```
y
```

```
Out[53]: 0      3.26
1      1.98
2      3.25
3      3.67
4      3.57
...
195    2.46
196    2.57
197    3.24
198    3.96
199    2.33
Name: package, Length: 200, dtype: float64
```

In [45]: `from sklearn.model_selection import train_test_split`In [72]: `X_train, X_test, y_train, y_test = train_test_split(X,y, test_size = 0.3, random_state = 2`In [73]: `from sklearn.linear_model import LinearRegression`In [74]: `lr2 = LinearRegression()`In [76]: `lr2.fit(X_train, y_train)`

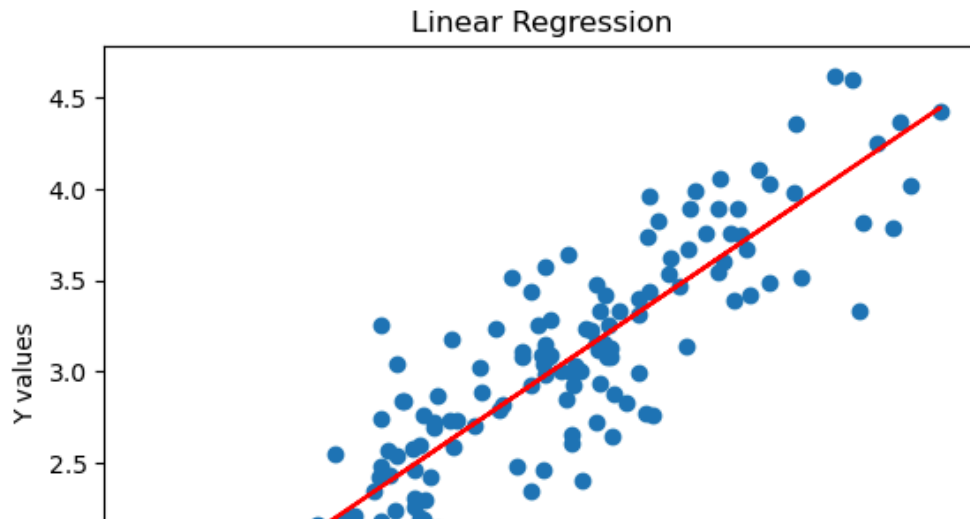
```
Out[76]: ▾ LinearRegression
LinearRegression()
```

In [77]: `lr2.coef_`Out[77]: `array([0.55574241])`In [78]: `lr2.intercept_`Out[78]: `-0.8781178361812025`

In [80]: `Y_predicted = lr2.predict(X_test)`

In [87]: `plt.scatter(X_train, y_train)`
`plt.plot(X_train, lr2.predict(X_train), color='red')`
`plt.xlabel("X values")`
`plt.ylabel("Y values")`
`plt.title("Linear Regression")`

Out[87]: `Text(0.5, 1.0, 'Linear Regression')`



In [88]: `mean_absolute_error(X_test, y_test)`

Out[88]: `4.007833333333333`

In [89]: `mean_squared_error(X_test, y_test)`

Out[89]: `16.357385`

In [90]: `r2_score(X_test, y_test)`

Out[90]: `-13.46488218445403`



Exp 3.ipynb × +

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```
[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
[4]: df = pd.read_csv("student_scores.csv")
df
```

```
[4]:
```

	Hours	Scores
0	2.5	21
1	5.1	47
2	3.2	27
3	8.5	75
4	3.5	30
5	1.5	20
6	9.2	88
7	5.5	60
8	8.3	81
9	2.7	25
10	7.7	85
11	5.9	62
12	4.5	41
13	3.3	42
14	1.1	17
15	8.9	95
16	2.5	30
17	1.9	24
18	6.1	67
19	7.4	69
20	2.7	30
21	4.8	54
22	3.8	35

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Exp 3.ipynb

        Code

247.886

```
[5]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25 entries, 0 to 24
Data columns (total 2 columns):
 #   Column  Non-Null Count  Dtype  
---  -
 0   Hours   25 non-null     float64
 1   Scores  25 non-null     int64  
dtypes: float64(1), int64(1)
memory usage: 528.0 bytes

[6]: X = df['Hours']
X = X.values.reshape(-1,1)
y = df['Scores']

[7]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, ran

[8]: lr = LinearRegression()
lr.fit(X_train, y_train)

[8]: ▼ LinearRegression ⓘ ?
      ► Parameters

[9]: lr.coef_

[9]: array([10.42786752])

[10]: lr.intercept_

[10]: np.float64(-0.4416515935882188)

[11]: y_pred = lr.predict(X_test)

[12]: plt.scatter(X_train, y_train)
plt.plot(X_train, lr.predict(X_train), color='red')
plt.xlabel("Study Hours")
plt.ylabel("Score")
```

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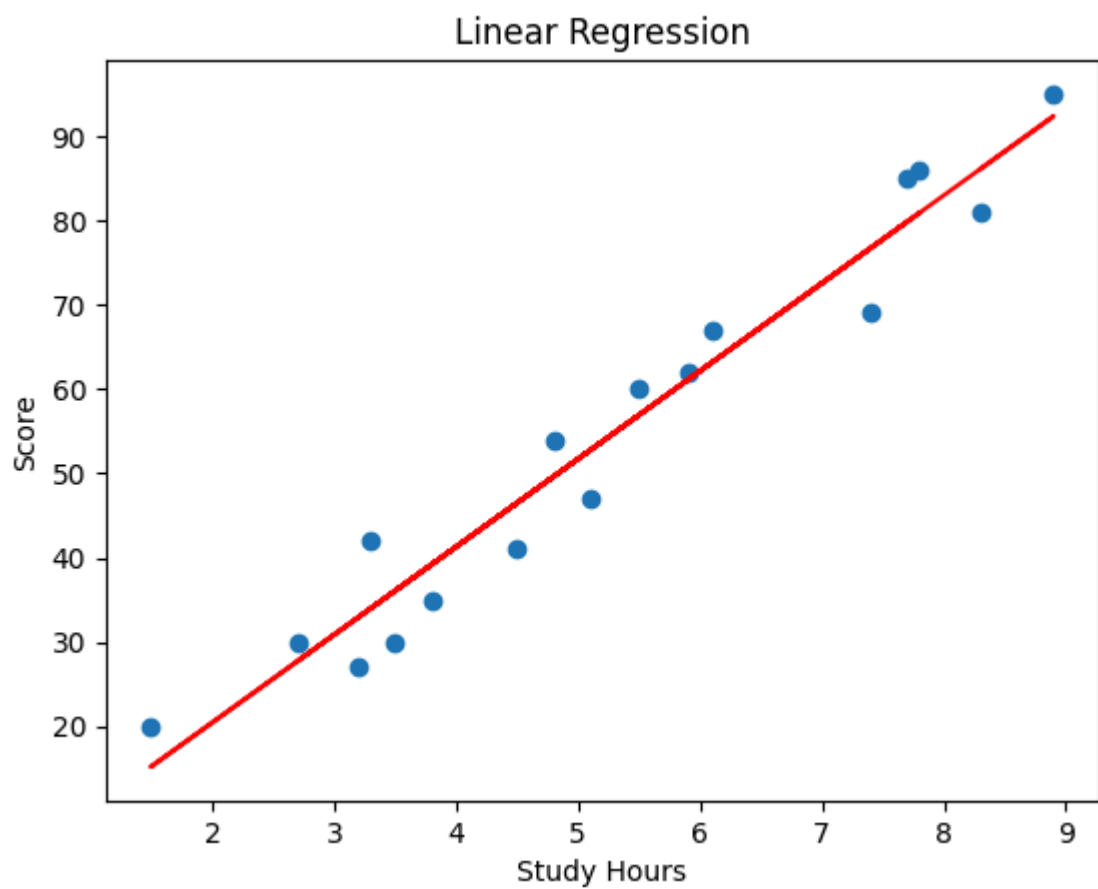
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```
[10]: lr.intercept_
```

```
[10]: np.float64(-0.4416515935882188)
```

```
[11]: y_pred = lr.predict(X_test)
```

```
[12]: plt.scatter(X_train, y_train)
plt.plot(X_train, lr.predict(X_train), color='red')
plt.xlabel("Study Hours")
plt.ylabel("Score")
plt.title("Linear Regression")
plt.show()
```



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